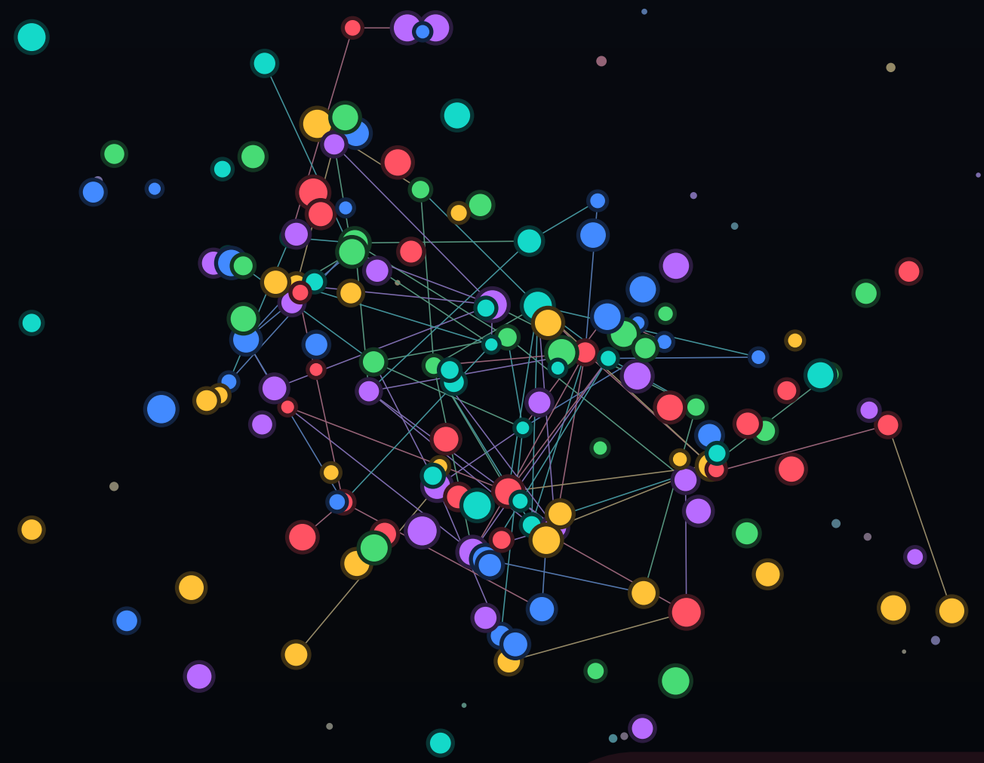
NEW DIRECTION

# Particle Lenia connects swarms to continuous lifeforms

It keeps the particle view from Particle Life, but borrows Lenia's smooth fields and growth functions.



thousands of local interactions

FIELD FIRST

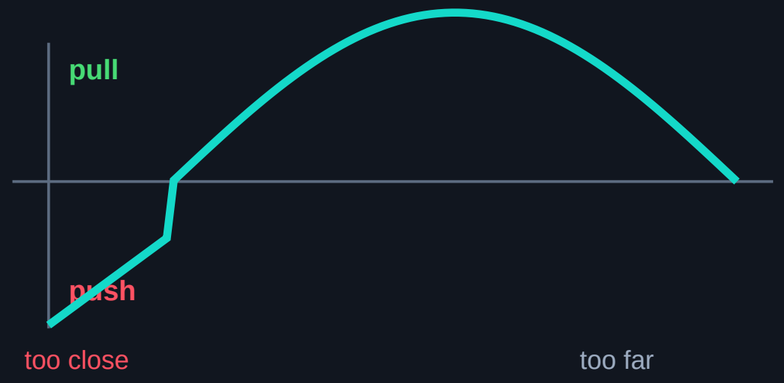
# Particles create a soft Lenia field

Every particle contributes a radial kernel. Other particles read the combined field where they are standing.

- The kernel is a ring, not a hard collision rule.
- The field is continuous, so motion changes smoothly.
- The same idea works in 2D and 3D.

## One pairwise force

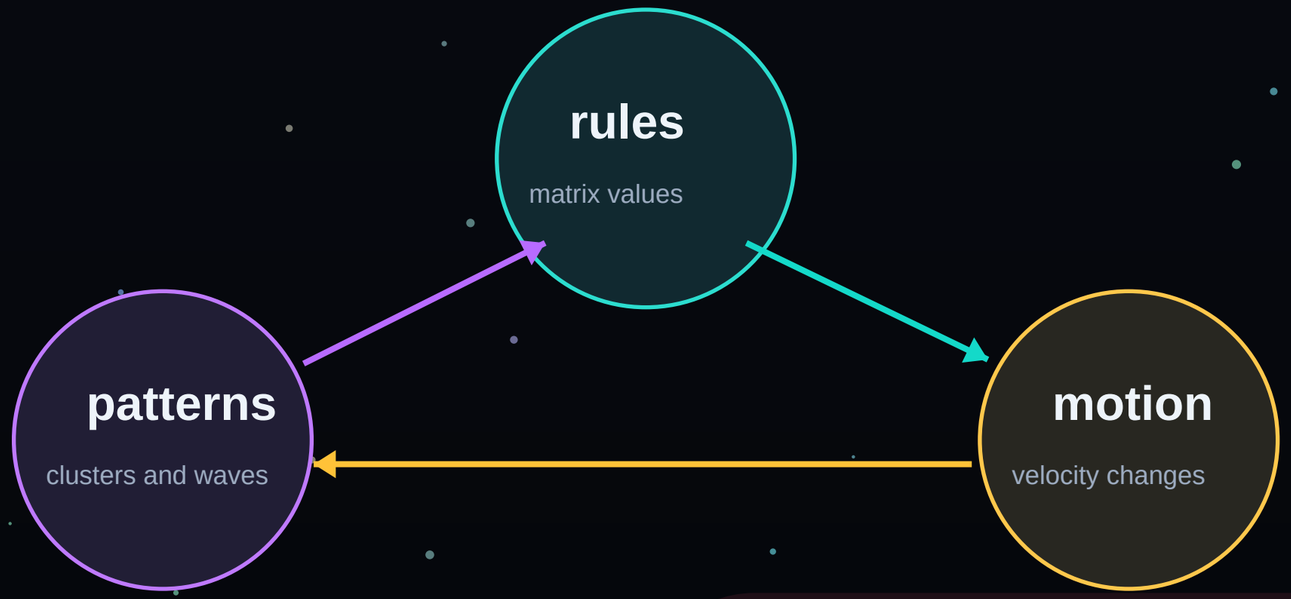
Distance changes how strongly particles react



ENERGY VIEW

# Motion becomes local energy descent

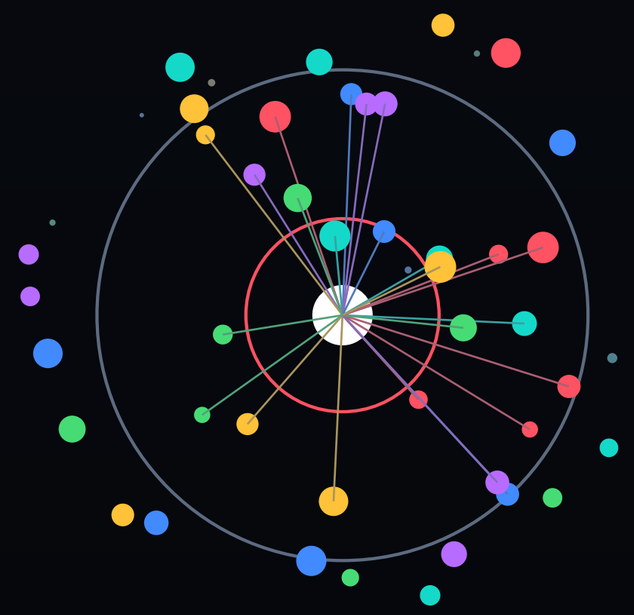
Particles seek places where the Lenia growth field is high while short-range repulsion prevents collapse.



MULTI-AGENT OPTIMIZATION

# Each particle optimizes its own local objective

That greedy local rule can escape the static minima that a single global optimizer would prefer.



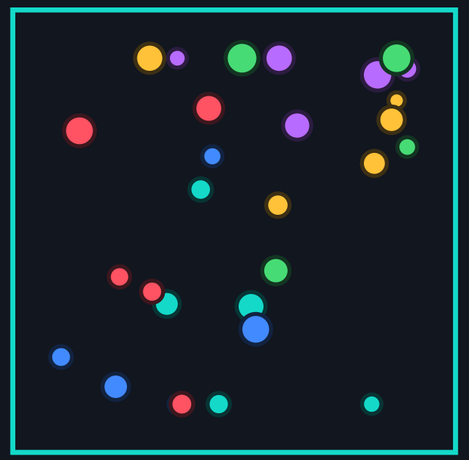
each particle only asks nearby questions

MASS CONSERVATION

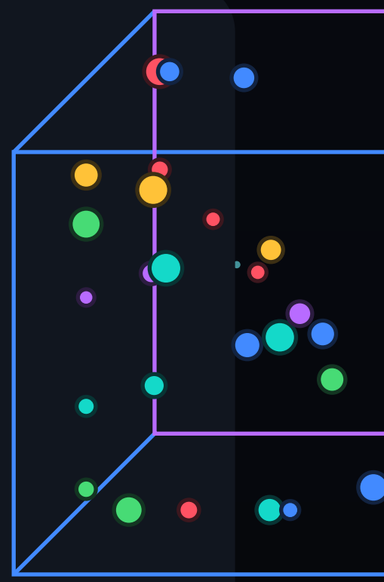
# Matter moves instead of appearing or vanishing

The particle count stays fixed. Complex forms emerge from rearrangement, not from creating new mass on a grid.

## 2D and 3D are the same idea



2D plane



3D volun

WHY IT MATTERS

# Particle Lenia is a bridge between a a d

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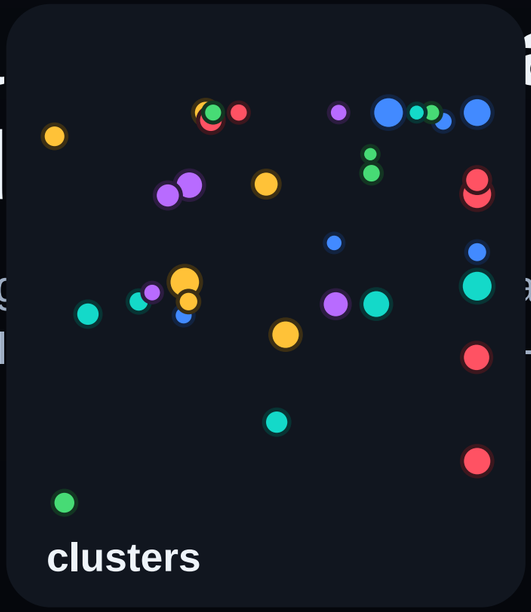
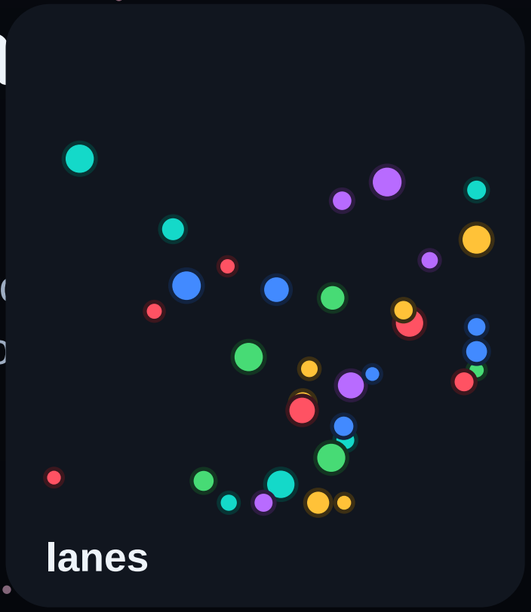
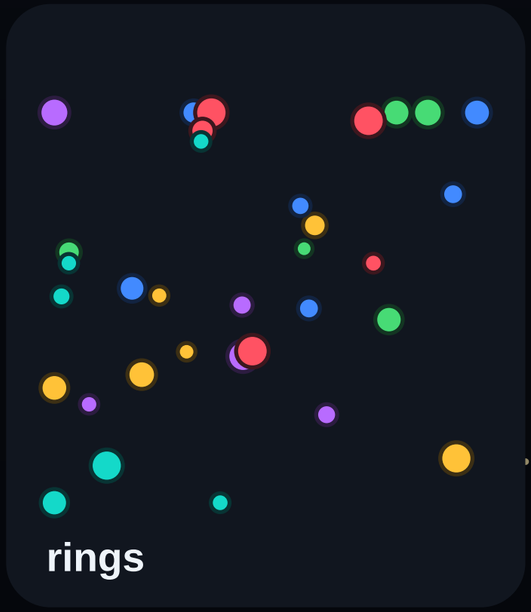
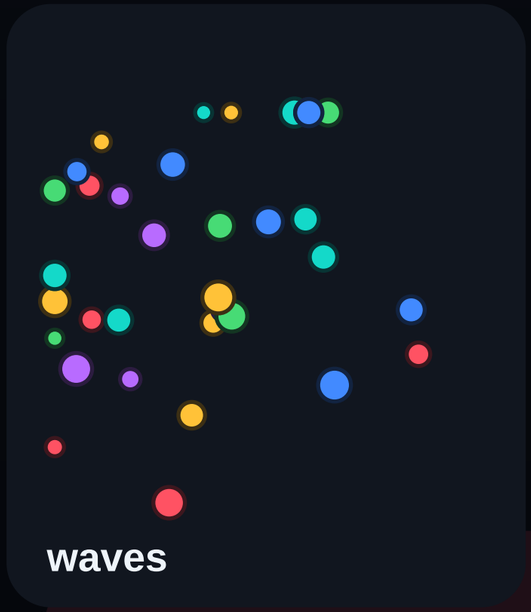

clusters


lanes


rings


waves